

Green functions

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1 Green's Functions

Format

This section gives a graduate-level introduction to Green's functions in electrostatics, including the derivation of the integral representation, the free-space kernel, and boundary-condition examples.

Green's Function for Poisson's Equation: Motivation

We want to solve Poisson's equation in electrostatics,

$$\nabla^2 V(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\epsilon_0},$$

either in free space or in a region with specified boundary conditions.

The key idea is to first solve the response to a *point source*. Define the Green's function $G(\mathbf{r}, \mathbf{r}')$ by

$$\nabla^2 G(\mathbf{r}, \mathbf{r}') = -\delta^{(3)}(\mathbf{r} - \mathbf{r}').$$

Here:

- $\mathbf{r}$ is the observation point,
- $\mathbf{r}'$ is the source point,
- $G(\mathbf{r}, \mathbf{r}')$ is the potential at $\mathbf{r}$ due to a unit point source at $\mathbf{r}'$.

Once G is known, the solution for an arbitrary charge density follows by superposition.

Main idea

Green's functions are inverse kernels of linear differential operators.

Green's Function for Poisson's Equation: First Step

Start from Poisson's equation written in the source coordinates $\mathbf{r}'$:

$$\nabla'^2 V(\mathbf{r}') = -\frac{\rho(\mathbf{r}')}{\varepsilon_0}.$$

Let the Green's function satisfy

$$\nabla'^2 G(\mathbf{r}, \mathbf{r}') = -\delta^{(3)}(\mathbf{r} - \mathbf{r}').$$

Multiply Poisson's equation by $G(\mathbf{r}, \mathbf{r}')$:

$$G(\mathbf{r}, \mathbf{r}') \nabla'^2 V(\mathbf{r}') = -\frac{G(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}')}{\varepsilon_0}.$$

Multiply the Green's-function equation by $V(\mathbf{r}')$:

$$V(\mathbf{r}') \nabla'^2 G(\mathbf{r}, \mathbf{r}') = -V(\mathbf{r}') \delta^{(3)}(\mathbf{r} - \mathbf{r}').$$

Subtract the second equation from the first:

$$G \nabla'^2 V - V \nabla'^2 G = -\frac{G \rho}{\varepsilon_0} + V(\mathbf{r}') \delta^{(3)}(\mathbf{r} - \mathbf{r}').$$

Green's Function for Poisson's Equation: Green's Identity

Use Green's second identity:

$$\nabla' \cdot (G \nabla' V - V \nabla' G) = G \nabla'^2 V - V \nabla'^2 G.$$

Therefore,

$$\nabla' \cdot (G \nabla' V - V \nabla' G) = -\frac{G\rho}{\varepsilon_0} + V(\mathbf{r}')\delta^{(3)}(\mathbf{r} - \mathbf{r}').$$

Now integrate over a volume $\mathcal{V}$:

$$\int_{\mathcal{V}} \nabla' \cdot (G \nabla' V - V \nabla' G) d^3 r' = -\frac{1}{\varepsilon_0} \int_{\mathcal{V}} G \rho d^3 r' + \int_{\mathcal{V}} V(\mathbf{r}') \delta^{(3)}(\mathbf{r} - \mathbf{r}') d^3 r'.$$

Apply the divergence theorem to the left-hand side:

$$\oint_{\partial \mathcal{V}} \left(G \frac{\partial V}{\partial n'} - V \frac{\partial G}{\partial n'} \right) da' = -\frac{1}{\varepsilon_0} \int_{\mathcal{V}} G \rho d^3 r' + \int_{\mathcal{V}} V(\mathbf{r}') \delta^{(3)}(\mathbf{r} - \mathbf{r}') d^3 r'.$$

Green's Function for Poisson's Equation: General Representation

Using the delta-function property,

$$\int_V V(\mathbf{r}') \delta^{(3)}(\mathbf{r} - \mathbf{r}') d^3r' = V(\mathbf{r}),$$

we obtain

$$\oint_{\partial V} \left(G \frac{\partial V}{\partial n'} - V \frac{\partial G}{\partial n'} \right) da' = -\frac{1}{\varepsilon_0} \int_V G(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}') d^3r' + V(\mathbf{r}).$$

Hence,

$$V(\mathbf{r}) = \frac{1}{\varepsilon_0} \int_V G(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}') d^3r' + \oint_{\partial V} \left(G \frac{\partial V}{\partial n'} - V \frac{\partial G}{\partial n'} \right) da'$$

Interpretation

The potential is determined by a volume contribution from the sources and a surface contribution from the boundary data.

Free-Space Green's Function: Symmetry Argument

In unbounded free space, the medium is translationally and rotationally invariant. Therefore the Green's function can only depend on the scalar distance

$$R = |\mathbf{r} - \mathbf{r}'|.$$

So we write

$$G(\mathbf{r}, \mathbf{r}') = G(R).$$

Away from the source point $R \neq 0$, the delta function vanishes, so

$$\nabla^2 G = 0.$$

For a spherically symmetric function of R ,

$$\nabla^2 G(R) = \frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{dG}{dR} \right).$$

Thus the equation becomes

$$\frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{dG}{dR} \right) = 0.$$

This is just Laplace's equation in spherical symmetry about the source point.

Free-Space Green's Function: Solving the ODE

Starting from

$$\frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{dG}{dR} \right) = 0,$$

multiply by R^2 :

$$\frac{d}{dR} \left(R^2 \frac{dG}{dR} \right) = 0.$$

Integrate once:

$$R^2 \frac{dG}{dR} = C_1.$$

Then

$$\frac{dG}{dR} = \frac{C_1}{R^2}.$$

Integrate again:

$$G(R) = \int \frac{C_1}{R^2} dR = -\frac{C_1}{R} + C_2.$$

Rename the constant:

$$G(R) = \frac{A}{R} + C_2.$$

To ensure the potential vanishes at infinity, we set

$$C_2 = 0.$$

Therefore the free-space Green's function must have the form

$$G(R) = \frac{A}{R}.$$

Free-Space Green's Function: Fixing the Constant

We now determine A from the defining equation

$$\nabla^2 G(\mathbf{r}, \mathbf{r}') = -\delta^{(3)}(\mathbf{r} - \mathbf{r}').$$

Integrate over a small sphere B_η of radius η centered at $\mathbf{r}'$:

$$\int_{B_\eta} \nabla^2 G d^3r = - \int_{B_\eta} \delta^{(3)}(\mathbf{r} - \mathbf{r}') d^3r = -1.$$

Apply the divergence theorem:

$$\oint_{S_\eta} \nabla G \cdot d\mathbf{a} = -1.$$

Since

$$G(R) = \frac{A}{R}, \quad \nabla G = \frac{d}{dR} \left(\frac{A}{R} \right) \hat{\mathbf{R}} = -\frac{A}{R^2} \hat{\mathbf{R}},$$

on the sphere $R = \eta$ we have

$$\nabla G \cdot d\mathbf{a} = -\frac{A}{\eta^2} da.$$

Free-Space Green's Function:

Therefore,

$$\oint_{S_\eta} \nabla G \cdot d\mathbf{a} = -\frac{A}{\eta^2} \oint_{S_\eta} da = -\frac{A}{\eta^2} (4\pi\eta^2) = -4\pi A.$$

Hence,

$$-4\pi A = -1 \quad \implies \quad A = \frac{1}{4\pi}.$$

Free-Space Green's Function: Final Result

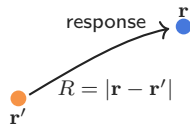
Substituting the normalization constant, we obtain

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|}$$

This is the free-space Green's function of the Laplacian in three dimensions.

It represents the potential at $\mathbf{r}$ due to a unit point source located at $\mathbf{r}'$.

The dependence on $|\mathbf{r} - \mathbf{r}'|^{-1}$ is exactly the Coulomb form, so the Coulomb kernel is the inverse kernel of the Laplace operator in free space.



Electrostatic Potential in Free Space

For free space, if the boundary term vanishes at infinity, the general representation reduces to

$$V(\mathbf{r}) = \frac{1}{\varepsilon_0} \int G(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}') d^3 r'.$$

Substitute

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|} :$$

$$V(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3 r'.$$

Thus,

$$V(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3 r'$$

Meaning

The total potential is the superposition of the contributions from all infinitesimal source elements $\rho(\mathbf{r}') d^3 r'$.

Example 1: Point Charge

Let the charge density be a point source at $\mathbf{r}_0$:

$$\rho(\mathbf{r}') = q \delta^{(3)}(\mathbf{r}' - \mathbf{r}_0).$$

Insert this into the Coulomb integral:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{q \delta^{(3)}(\mathbf{r}' - \mathbf{r}_0)}{|\mathbf{r} - \mathbf{r}'|} d^3r'.$$

Using the delta-function identity

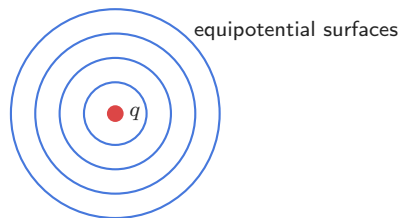
$$\int f(\mathbf{r}') \delta^{(3)}(\mathbf{r}' - \mathbf{r}_0) d^3r' = f(\mathbf{r}_0),$$

we obtain

$$V(\mathbf{r}) = \frac{q}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_0|}.$$

Therefore,

$$V(\mathbf{r}) = \frac{q}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_0|}$$



Dirichlet Green's Function

Suppose the potential is specified on the boundary:

$$V(\mathbf{r}') \text{ known on } \partial\mathcal{V}.$$

To adapt the integral formula to this case, choose a Green's function satisfying the Dirichlet condition

$$G(\mathbf{r}, \mathbf{r}') = 0 \quad \text{for } \mathbf{r}' \in \partial\mathcal{V}.$$

Then the term involving $G \partial V / \partial n'$ vanishes on the boundary, and the solution becomes

$$V(\mathbf{r}) = \frac{1}{\varepsilon_0} \int_{\mathcal{V}} G(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}') d^3r' - \oint_{\partial\mathcal{V}} V(\mathbf{r}') \frac{\partial G}{\partial n'} da'$$

Why this is useful

The boundary condition is built directly into the Green's function itself.

Example 2: Grounded Conducting Plane

For a grounded conducting plane,

$$V = 0 \quad \text{on the plane.}$$

This is a Dirichlet boundary condition. The appropriate Green's function can be constructed with the method of images.

If the real source is at $\mathbf{r}'$ and its mirror point is $\mathbf{r}''$, define

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|} - \frac{1}{4\pi|\mathbf{r} - \mathbf{r}''|}$$

On the plane, the two distances are equal:

$$|\mathbf{r} - \mathbf{r}'| = |\mathbf{r} - \mathbf{r}''|,$$

so the two terms cancel and therefore

$$G = 0.$$

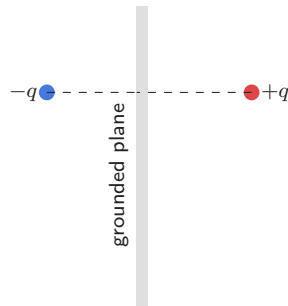
Hence the boundary condition is satisfied exactly.

Method of Images: Geometric Picture

The image source is not physical. It is a mathematical construction that modifies the kernel so that the potential vanishes on the conducting boundary.

For a real point charge $+q$ near a grounded plane:

- the real source is located above the plane,
- an image source $-q$ is placed symmetrically below the plane,
- the sum of their contributions produces $V = 0$ on the plane.



Interpretation

The image term changes the free-space kernel into a boundary-adapted Green's function.

Why Green's Functions Matter

- Green's functions solve the response to a point source.
- By linearity and superposition, they generate the solution for arbitrary $\rho(\mathbf{r})$.
- The operator determines the local form of the kernel.
- The geometry and boundary conditions determine the global form of the kernel.

For electrostatics in free space,

$$G(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|}$$

and therefore

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'$$

Graduate-level takeaway

A Green's function is the inverse kernel of a linear differential operator, and it encodes both source propagation and boundary geometry in a single object.